Day 5: Prime or composite? Prove it.

Name _____ Date _____

I can tell if a number is prime or composite, and prove it.

Only one rectangle? Which numbers from 1 to 20 make only one rectangle? Sketch on grid paper. Write the numbers here.

Only one rectangle: _____

Words. Prime: exactly ____ factors. Composite: _____ than two factors. 1 is _____.

Example 1. Is 31 prime or composite? Prove it.

____ is _____: I tried ___ to ___ and found no pair.

Example 2. Is 51 prime or composite? Prove it.

____ is _____, because ___ × ___ = ____.

Sort. Write each number after its word. Write a pair for each composite. For each prime, write the last number you tried.

1. Sort 23, 33 and 85.

   Prime, last number tried: _____ Neither: _____

   Composite, with its pair: _____

2. Sort 1, 37 and 69.

   Prime, last number tried: _____ Neither: _____

   Composite, with its pair: _____

3. Sort 2, 63 and 89.

   Prime, last number tried: _____ Neither: _____

   Composite, with its pair: _____

4. Ring every prime: 41, 42, 43, 44, 45, 46, 47, 48, 49. Write a pair for each odd number you did not ring.

   Pairs: _____

5. Ring every prime on your 1 to 30 chart. Write a proof pair under each odd composite.

Fix it. The answer is wrong. Say what went wrong. Then do it right.

6. Kai says: "87 is prime. It is odd, and it is not in the times tables."

   What went wrong: _____

7. Pia says: "1 is prime. 2 is composite, because it is even."

   What went wrong: _____

Write to Ben.

8. Ben says 91 is prime. Write to Ben. Is he right? Prove it.

Challenge. Can a multiple of 7 be prime? Explain.

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